No chemical studies have so far been performed on meitnerium, darmstadtium or roentgenium.

The chemical behaviour of copernicium has been studied in gas phase experiments where Cn has been allowed to adsorb on surfaces in a system with alpha detectors at different temperature. The temperature gradient goes from room temperature to the temperature of liquid nitrogen (−196 °C). The results imply that Cn is very volatile but it still has the adsorption characteristics of a metal. It is therefore assumed that it belongs to group 12.

No chemical studies have so far been performed on element 113.

The chemical behaviour of element 114 has been studied in gas phase experiments where 114 has been allowed to adsorb in a system with alpha detectors at different temperature. The temperature gradient goes from room temperature to the temperature of liquid nitrogen (−196 °C). The results imply that 114 is very volatile (even more volatile than Cn) but it still has the adsorption characteristics of a metal. It is therefore assumed that it belongs to group 14. Cn and 114 can both be characterized as "gaseous metals". Cn and 114 are the first elements where relativistic effects significantly alter the chemical properties.

No chemical studies have so far been performed on elements 115, 116, 117 and 118.

It should be emphasized that many of the results in this region are based on very few decay events, in some cases only two atoms, due to the low production rates and short half-lives. This gives large uncertainties and there is room for surprises in future experiments. The only trtansactinide elements where a significant amount of data has been collected with different methods in independent experiments are Rf and Db.

14.6. THE END OF THE PERIODIC TABLE

A question that arises in heavy element research is: how far can we reach? Is there a limit on the number of elements that cannot be exceeded? The answer is that theoretically it is possible to find quite a number of new elements above $Z = 118$ but in practice the number of elements will be limited by experimental difficulties. To reach far beyond $Z = 125$ and approach $N = 184$ will require even more neutron-rich actinide targets and more neutron-rich projectiles, e.g. short-lived fission products. The limit of hot fusion may thus be reached short of arriving at the proposed magic region around $N = 184$.

14.7. EXERCISES

14.1. What nuclear reactions would be suitable to make gold?

14.2. What fraction of neptunium is in the +4 state in a 0.1 M $Fe(SO_4)_2$ solution of acidity 1 M H^+, assuming that Np initially is present as Np(V) at a total concentration of 9.4 mM?

14.3. Irradiation of ^{238}U with deuterium yields ^{238}Pu. Will any other plutonium isotope be produced?

14.4. What are the decay products of ^{252}Cf, of ^{253}Cf and of ^{254}Cf?

14.5. ^{244}Pu decays through spontaneous fission with a half-life of 6.6×10^{10} y. Estimate the number of neutrons emitted per fission if the measured n-emission rate is 1890 n s^{-1} g^{-1}.

14.6. Flerov bombarded ^{207}Pb with ^{54}Cr and obtained a product which within 4-10 ms decayed by spontaneous fission. Suggest a product nucleus.

14.7. In Figure 14.9, Eu(III) is more easily extracted than its homolog Am(III). Suggest an explanation.

14.8. A cardiac pacemaker contained 150 mg of ^{238}Pu. What is its heat output? Use data in the isotope chart.

14.9. What electronic configuration would you ascribe to nobelium considering its place in the periodic system and its chemistry?

14.10. Based on the redox diagrams in Figure 14.8, estimate the reduction potentials for the one-electron step in the following series of plutonium species in acid solution: Pu(III), Pu(IV), Pu(V), and Pu(VI).

14.8. LITERATURE

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